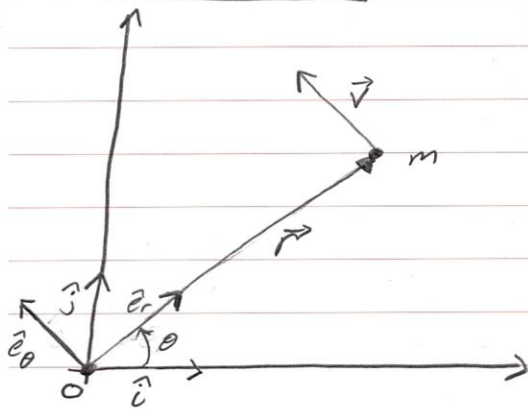
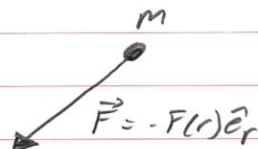


#7) a) Find a central Force Law $F(r)$ so that, comparing circular orbits of varying radii, the speed v is independent of radius.

Coordinate System



FBD



Transformation Matrix

$$\begin{bmatrix} \hat{e}_r \\ \hat{e}_\theta \end{bmatrix} = \begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix} \begin{bmatrix} \hat{i} \\ \hat{j} \end{bmatrix}$$

Kinematics

$$\vec{r} = r \hat{e}_r \quad \rightarrow \text{length of } r \text{ is fixed}$$

$$\dot{\vec{r}} = \dot{\vec{v}} = \dot{r} \hat{e}_r + r \dot{\theta} \hat{e}_\theta$$

$$\vec{v} = r \dot{\theta} \hat{e}_\theta$$

Find $\dot{\theta}$ so that $|v|$ is indep. of radius

$$\boxed{\dot{\theta} = \frac{|v|}{r}} \quad (1)$$

Find $\vec{a}$ then find $\vec{F}$

$$\vec{v} = \vec{a} = \cancel{\dot{r} \hat{e}_r} + r \ddot{\theta} \hat{e}_\theta + r \dot{\theta}^2 \hat{e}_r$$

$= 0$ $\because r$ is constant.

$$\vec{F} = m \vec{a}$$

$$\vec{F}(r) \hat{e}_r = (r \ddot{\theta} \hat{e}_\theta - r \dot{\theta}^2 \hat{e}_r) m$$

Dot w/ $\hat{e}_r$

$$F(r) = m r \dot{\theta}^2$$

Dot w/ $\hat{e}_\theta$

$$\ddot{\theta} = 0$$

plug in $\dot{\theta}$ found above (1)

$$\boxed{F(r) = \frac{m |v|^2}{r}}$$

b) Find a periodic motion for a central force other than
 $F = -kr$ or $F = -\frac{GMm}{r^2}$

$$\text{Try } \vec{F}(r) = \left(-\frac{GMm}{r^2} - kr \right) \hat{e}_r$$

Find EOM's

$$\vec{r} = r \hat{e}_r$$

$$\vec{v} = \dot{r} \hat{e}_r + r \dot{\theta} \hat{e}_\theta$$

$$\begin{aligned} \vec{a} &= \ddot{r} \hat{e}_r + \dot{r} \dot{\theta} \hat{e}_\theta + \dot{r} \dot{\theta} \hat{e}_\theta + r \ddot{\theta} \hat{e}_\theta - r \dot{\theta}^2 \hat{e}_r \\ &= (\ddot{r} - r \dot{\theta}^2) \hat{e}_r + (r \ddot{\theta} + 2\dot{r} \dot{\theta}) \hat{e}_\theta \end{aligned}$$

$$\vec{F} = m \vec{a}$$

$$\left(-\frac{GMm}{r^2} - kr \right) \hat{e}_r = m(\ddot{r} - r \dot{\theta}^2) \hat{e}_r + m(r \ddot{\theta} + 2\dot{r} \dot{\theta}) \hat{e}_\theta$$

Dot w/ $\hat{e}_r$

$$m\ddot{r} - mr\dot{\theta}^2 = -\frac{GMm}{r^2} - kr$$

$$\boxed{\ddot{r} = -\frac{GM}{r^2} - \frac{k}{m}r + r\dot{\theta}^2}$$

Dot w/ $\hat{e}_\theta$

$$0 = r\ddot{\theta} + 2\dot{r}\dot{\theta}$$

$$\boxed{\ddot{\theta} = -\frac{2\dot{r}\dot{\theta}}{r}}$$

Set up state space

$$z = \begin{bmatrix} r \\ \dot{r} \\ \dot{\theta} \\ \ddot{\theta} \end{bmatrix} \quad \dot{z} = \begin{bmatrix} \dot{r} \\ \ddot{r} \\ \dot{\theta} \\ \ddot{\theta} \end{bmatrix}$$

Numerically Integrate in Matlab

Motions always have regular patterns.

```

function mae5730p7b()

close all
clear all
clc

%Assign parameters
p.G = 3;
p.M = 1;
p.m = 2;
p.k = 609;

%Time of integration
T = 1.596;
tspan = linspace(0,T,1001);

%I.C's
%z0 = [r0 rdot0 theta0 thetadot0]
z0 = [1 0 0 1];

%Solve
[t,zarray] = ode45(@rhs,tspan,z0,[],p);

%Turn r and theta into cartesian coordinates for plotting
r = zarray(:,1);
x = r.*cos(zarray(:,3));
y = r.*sin(zarray(:,3));

%check final conditions against initial conditions
zcheck = zarray(end,:);
turns = zarray(end,3)/(2*pi);
turns = floor(turns);
zcheck(3) = (zcheck(3)/(turns))-(2*pi);
check = zcheck-z0;
disp('Differnces in - r, rdot, theta, thetadot')
disp(check)

%plot
figure
hold on
%path
plot(x,y,'LineWidth',2)
%origin
plot(0,0,'k+')
axis equal
xlabel('X position','FontSize',16)
ylabel('Y position','FontSize',16)
title('Periodic Motion of Central Force  $F=-GmM/r^2 - kr$ ','FontSize',16)
hold off

% Other motion
%Time of integration
T = 5;
tspan = linspace(0,T,1001*T);

```



```

%I.C's
%z0 = [r0 rdot0 theta0 thetadot0]
z0 = [1 10 0 -1];

%set tolerances
% RelTol = 1e-1;
% AbsTol = 1e-1;
% odeset = [RelTol AbsTol];

%Solve
[t,zarray] = ode45(@rhs,tspan,z0,[],p);

%Turn r and theta into cartesian coordinates for plotting
r = zarray(:,1);
x = r.*cos(zarray(:,3));
y = r.*sin(zarray(:,3));

%check final conditions against initial conditions
zcheck = zarray(end,:);
%subtract out largest whole number of turns
turns = zarray(end,3)/(2*pi);
turns = floor(turns);
zcheck(3) = (zcheck(3)/(turns))-(2*pi);
%diffrence between final and initial conditions
check = zcheck-z0;
% disp('Differnces in - r, rdot, theta, thetadot')
% disp(check)

%plot
figure
hold on
%path
plot(x,y,'LineWidth',2)
%origin
plot(0,0,'k+')
axis equal
xlabel('X position','FontSize',16)
ylabel('Y position','FontSize',16)
title('Other Motion of Central Force  $F=-GmM/r^2 - kr$ ','FontSize',16)
hold off
end

```

```

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

```

```

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
function zdot =rhs(t,z,p)
%z = [r rdot theta thetadot]
%zdot = [rdot rdoubledot thetadot thetadoubledot]

```

```
G = p.G;
M = p.M;
m = p.m;
k = p.k;
```

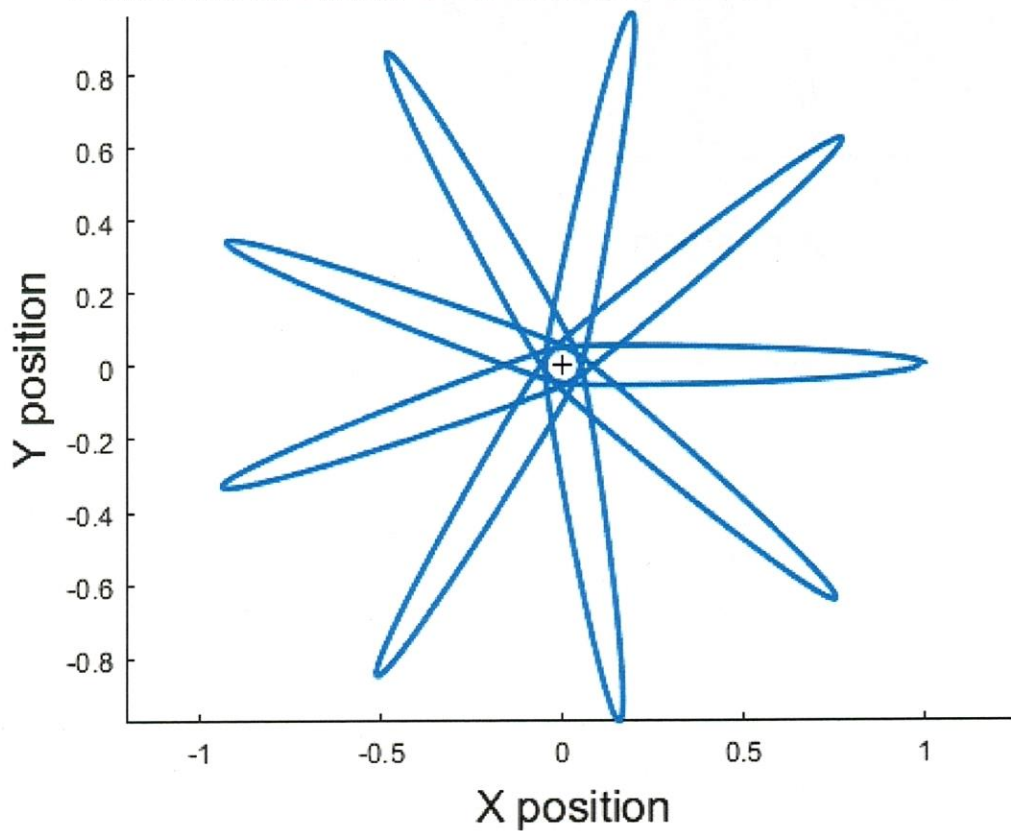
```
zdot = [z(2); (-G*M/z(1)^2) - (k*z(1)/m) + (z(1)*z(4)^2); z(4); -2*z(2)*z(4)/z(1)];
```

```
end
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

```
Differences in - r, rdot, theta, thetadot
-0.0166    0.0010    0.0002   -0.0123
```

Periodic Motion of Central Force $F = -GmM/r^2 - kr$



Other Motion of Central Force $F = -GmM/r^2 - kr$

